Lecture 11

Scattering states: transmission, reflection, tunnelling

Objectives. *For the compactly supported potentials of Lecture 10, scattering solutions have amplitudes r and t that satisfy $|r|^2 + |t|^2 = 1$. They are not states. Packets are. We compute amplitudes for a step and for a barrier, by matching. For the δ Hamiltonian, we prove that it is self-adjoint and find its level and its transmission. We build the evolution of packets from the free one, without the spectral theorem. Wave operators connect packets to the stationary amplitudes.*

- Compute reflection and transmission from the current.
- Explain tunnelling through a finite barrier.
- State the potential and momentum-support hypotheses of the packet transmission limit.

Lecture 11, first part

Stationary solutions and their currents

For a potential that vanishes outside a bounded interval, every eigenvalue is negative (Theorem 10.7). Above 0 the equation (10.1) still has solutions, but they do not decay, so they are not eigenvectors. In this lecture we read them as streams of particles. A wave e^{ikx} comes in from the left. Part of it is reflected and part is transmitted, with amplitudes r and t. We first show that such solutions exist.

11.1 Waves that do not decay

Let V be a potential that vanishes outside $[-R, R]$ (Definition 10.1), and let $E = \hbar^2k^2/2m$ with $k > 0$. Outside $[-R, R]$ the equation (10.1) reads $\psi'' = -k^2\psi$, and its solutions there are the combinations of e^{ikx} and e^{-ikx} (§ 10.2), waves of momenta $\hbar k$ and $-\hbar k$. Lecture 10 solved the equation only where it could write the solutions down. In general they exist.

Lemma 11.1 (Existence). *Let V be a potential, E, x_0 real and $a, b \in \mathbb{C}$. There is exactly one continuously differentiable function ψ on $\mathbb{R}$ that is twice differentiable where V is continuous, satisfies (10.1) there, and has $\psi(x_0) = a$ and $\psi'(x_0) = b$.*

Proof. Uniqueness is Lemma 10.5. Put $g = (2m/\hbar^2)(V - E)$ and $C = \sup|g|$, let $\psi_0(x) = a + b(x - x_0)$, and

$$\psi_{n+1}(x) = a + b(x - x_0) + \int_{x_0}^{x}\int_{x_0}^{s} g(y)\,\psi_n(y)\,\mathrm{d}y\,\mathrm{d}s.$$

Fix $L > 0$, and let M be the largest value of $|\psi_1 - \psi_0|$ on $I = [x_0 - L, x_0 + L]$. Since the double integral of $|y - x_0|^{2n}$ is $|x - x_0|^{2n+2}/(2n+1)(2n+2)$, induction gives $|\psi_{n+1} - \psi_n| \le MC^n |x - x_0|^{2n}/(2n)!$ on I. Since $\sum_n MC^n L^{2n}/(2n)! = M\cosh(\sqrt{C}\,L)$ is finite, the ψ_n converge uniformly on I, and their limit ψ satisfies the same equation without the index. The inner integral is continuous in s, so ψ is continuously differentiable, with $\psi'(x) = b + \int_{x_0}^{x} g\psi\,\mathrm{d}y$, and $\psi'' = g\psi$ where V is continuous. This is (10.1), and $\psi(x_0) = a$, $\psi'(x_0) = b$. Since L is arbitrary, ψ is defined on $\mathbb{R}$. ∎

Proposition 11.2 (Scattering solutions). *Let V be a potential that vanishes outside $[-R, R]$, and let $k > 0$. There is exactly one solution ψ_k of (10.1) with $E = \hbar^2k^2/2m$, as in Lemma 11.1, such that for some numbers r and t*

$$\psi_k(x) = e^{\mathrm{i}kx} + r\,e^{-\mathrm{i}kx} \quad \text{for } x \le -R, \qquad \psi_k(x) = t\,e^{\mathrm{i}kx} \quad \text{for } x \ge R. \tag{11.1}$$

They satisfy $|r|^2 + |t|^2 = 1$, and $t \neq 0$.

Proof. Let φ be the solution with $\varphi(R) = e^{\mathrm{i}kR}$ and $\varphi'(R) = \mathrm{i}ke^{\mathrm{i}kR}$. By uniqueness, $\varphi = e^{\mathrm{i}kx}$ for $x \ge R$, and $\varphi = ae^{\mathrm{i}kx} + be^{-\mathrm{i}kx}$ for $x \le -R$, for some a and b. Since V and E are real, $\overline{\varphi}$ is a solution too, and as in Theorem 10.7 the Wronskian $W = \overline{\varphi}\varphi' - \overline{\varphi}'\varphi$ is constant. It is $2\mathrm{i}k$ for $x \ge R$ and $2\mathrm{i}k(|a|^2 - |b|^2)$ for $x \le -R$. So $|a|^2 - |b|^2 = 1$, and $a \neq 0$. Then $\psi_k = \varphi/a$ has the form (11.1) with $t = 1/a$ and $r = b/a$, and $|r|^2 + |t|^2 = (|b|^2 + 1)/|a|^2 = 1$. If $\hat{\psi}$ is a second such solution, with $\hat{r}$ and $\hat{t}$, then $d = \psi_k - \hat{\psi}$ is $(t - \hat{t})e^{\mathrm{i}kx}$ on the right and $(r - \hat{r})e^{-\mathrm{i}kx}$ on the left, and its Wronskian $\overline{d}d' - \overline{d}'d$ is $2\mathrm{i}k\,|t - \hat{t}|^2$ on the right and $-2\mathrm{i}k\,|r - \hat{r}|^2$ on the left. So both vanish, $d = 0$ for $x \ge R$, and $d = 0$ by uniqueness. ∎

The numbers $r = r(k)$ and $t = t(k)$ are the *reflection* and *transmission amplitudes*. Since $|\psi_k| = |t| > 0$ for $x \ge R$, ψ_k is not square integrable, as Theorem 10.7 requires. So it is not a state. Cut off far to the right, it gives, up to the factor t, the Weyl sequences of the proof of Theorem 10.10, with $p_0 = \hbar k$. What it describes is read from its *current* $j = (\hbar/m)\,\mathrm{Im}(\overline{\psi}\psi')$. This is the formula (2.5) of the probability current, applied here to a function that is not a state. The current is $(\hbar/2m\mathrm{i})W$ and so constant. It is $(\hbar k/m)(1 - |r|^2)$ on the left and $(\hbar k/m)\,|t|^2$ on the right. Read $e^{\mathrm{i}kx}$ as a stream of particles of velocity $\hbar k/m$. Then, of the incoming current, the fraction $|r|^2$ is reflected and $|t|^2$ is transmitted. § 11.5 says what this means for a state.

Self-check 11.1

1. Show that $|\psi_k(x)|^2 = 1 + |r|^2 + 2\,\mathrm{Re}(re^{-2\mathrm{i}kx})$ for $x \le -R$, and find its largest and smallest values.

11.2 The step

A step, which does not vanish on one side, changes the wave number there. Let $V_0 > 0$, and let $V = V_0$ on $(0, \infty)$ and $V = 0$ on $(-\infty, 0]$. It is bounded and real, so the proof of Proposition 10.2 shows that $H_0 + V$ is self-adjoint on $D(H_0)$. The proofs of Lemmas 10.5 and 11.1 use only that V is real, bounded and continuous except at finitely many jumps, so they apply too.

For $E > V_0$ let $k = \sqrt{2mE}/\hbar$ and $k' = \sqrt{2m(E - V_0)}/\hbar$, and match $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ for $x < 0$ to $te^{\mathrm{i}k'x}$ for $x > 0$. By Lemma 11.1, which applies here, a solution is continuously differentiable, so the values and the slopes of the two pieces agree at 0:

$$1 + r = t \quad \text{and} \quad k(1 - r) = k't.$$

Solving these two equations gives

$$t = \frac{2k}{k+k'}, \qquad r = \frac{k-k'}{k+k'}.$$

The current is $(\hbar k/m)(1-|r|^2)$ on the left and $(\hbar k'/m)\,|t|^2$ on the right, since the transmitted particles move with the velocity $\hbar k'/m$. So the reflected and transmitted fractions of the current are

$$R = \left(\frac{k-k'}{k+k'}\right)^2, \qquad T = \frac{k'}{k}\,|t|^2 = \frac{4kk'}{(k+k')^2}, \qquad R+T=1.$$

The transmitted fraction is not $|t|^2$, since the current weights the amplitude by the velocity. At $E = 2V_0$, for example, $k'/k = 1/\sqrt{2}$, so $t \approx 1.172$ and $T \approx 0.971$. The density on the right, $|t|^2 \approx 1.37$, exceeds the incoming density 1, yet the transmitted current is smaller than the incoming one, because the particles on the right are slower.

For $0 < E < V_0$, the wave on the right is a multiple of e^{-qx}, with $q = \sqrt{2m(V_0-E)}/\hbar$. Its current is 0, so the current $(\hbar k/m)(1-|r|^2)$ on the left vanishes too, and $|r| = 1$, $R = 1$. The same matching gives the amplitudes,

$$t = \frac{2k}{k+\mathrm{i}q}, \qquad r = \frac{k-\mathrm{i}q}{k+\mathrm{i}q}.$$

Everything is reflected. Yet the wave does not vanish for $x > 0$. It decays over the length $1/q$, and a packet enters the step for a while before it returns. Figure 11.1 draws both cases.

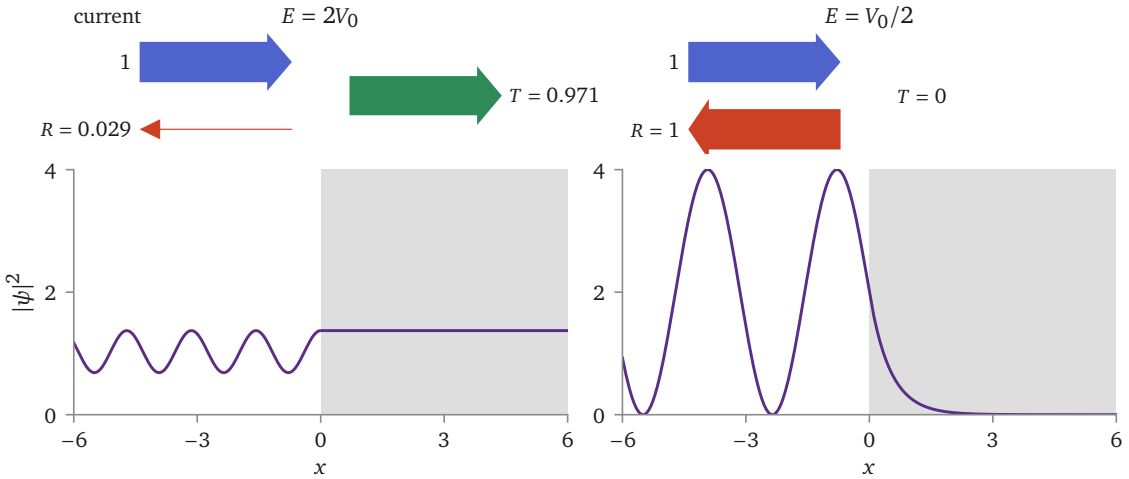


Figure 11.1: The step in units with $\hbar = m = V_0 = 1$, at $E = 2V_0$ (left) and at $E = V_0/2$ (right). Upper strips: arrows whose widths are proportional to the current, as fractions of the incoming current. The transmitted arrow is absent at $E = V_0/2$, where $T = 0$. Lower panels: the density $|\psi|^2$ of the matched solution, with the region of the step grey.

This section is stationary. § 11.5 gives such fractions their meaning for packets, for potentials that vanish outside a bounded interval. The step is not one of them.

Self-check 11.2

1. Show that $T \to 0$ as E falls to V_0, and $T \to 1$ as $E \to \infty$.
2. A particle meets the step with $E = 4V_0/3$. Write the two matching equations, and find r, t, R

and T. Check that $R + T = 1$ and that $|r|^2 + |t|^2 \neq 1$.

11.3 Tunnelling

A barrier of height $V_0 > 0$ and full width ℓ is the potential $V = V_0$ on $(0, \ell)$ and $V = 0$ outside. A classical particle with energy $E < V_0$ is reflected. We match a stationary solution at both edges, as for the step.

Let $0 < E < V_0$, and put $k = \sqrt{2mE}/\hbar$ and $q = \sqrt{2m(V_0 - E)}/\hbar$. Inside the barrier the equation reads $\psi'' = q^2\psi$. So the solution has the form

$$\psi(x) = \begin{cases} e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}, & x \leq 0, \\ Ae^{qx} + Be^{-qx}, & 0 \leq x \leq \ell, \\ te^{\mathrm{i}kx}, & x \geq \ell. \end{cases}$$

The barrier is a potential, so by Lemma 11.1 the solution is continuously differentiable. So the values and the slopes of the pieces agree at the two edges. At 0 this gives

$$1 + r = A + B, \qquad \mathrm{i}k(1 - r) = q(A - B),$$

and at ℓ it gives

$$Ae^{q\ell} + Be^{-q\ell} = te^{\mathrm{i}k\ell}, \qquad q\big(Ae^{q\ell} - Be^{-q\ell}\big) = \mathrm{i}kte^{\mathrm{i}k\ell}.$$

Figure 11.2 draws the solution of these four equations for $\hbar = m = V_0 = 1$, $\ell = 2$ and $E = 0.6$, where $k \approx 1.095$ and $q \approx 0.894$. The density inside the barrier is not a pure exponential, since the solution combines a growing and a decaying exponential. At $x = \ell$ it is about 0.102, and the decaying part alone gives about 0.064.

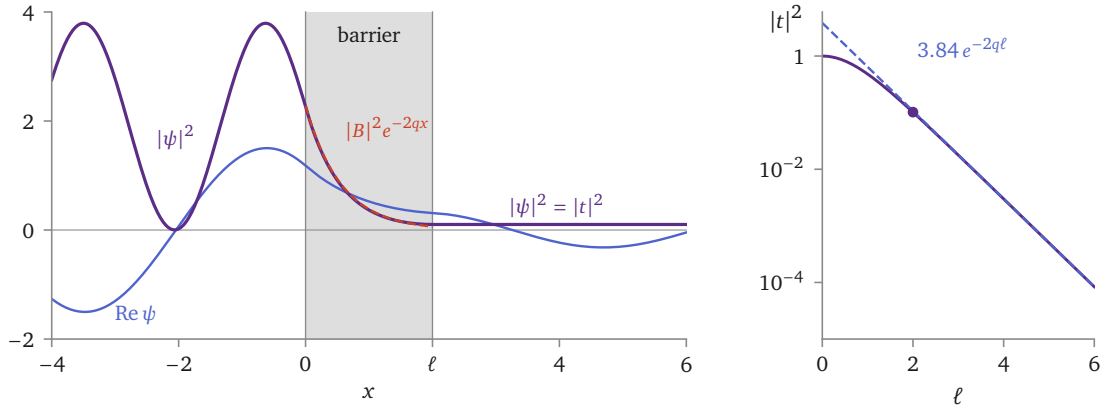


Figure 11.2: The stationary solution at a barrier, in units with $\hbar = m = V_0 = 1$, at $E = 0.6$. Left, for $\ell = 2$: $\mathrm{Re}\,\psi$ and $|\psi|^2$ from the four matching equations, with the barrier grey; dashed, the density $|B|^2\,e^{-2qx}$ of the decaying part alone. Right: $|t|^2$ against ℓ on a logarithmic scale, with the approximation $3.84\,e^{-2q\ell}$ dashed; the dot marks $\ell = 2$.

We eliminate A, B and r in three steps. First, the equations at ℓ give

$$Ae^{q\ell} = \tfrac{1}{2}(1 + \mathrm{i}k/q)\,te^{\mathrm{i}k\ell}, \qquad Be^{-q\ell} = \tfrac{1}{2}(1 - \mathrm{i}k/q)\,te^{\mathrm{i}k\ell}.$$

Next, $\mathrm{i}k$ times the first equation at 0, added to the second, removes r:

$$2\mathrm{i}k = (q + \mathrm{i}k)A - (q - \mathrm{i}k)B.$$

Finally, we insert A and B and expand the two squares:

$$2\mathrm{i}k = \frac{te^{\mathrm{i}k\ell}}{2q}\left((q+\mathrm{i}k)^2 e^{-q\ell} - (q-\mathrm{i}k)^2 e^{q\ell}\right) = te^{\mathrm{i}k\ell}\left(2\mathrm{i}k\cosh(q\ell) - \frac{q^2-k^2}{q}\sinh(q\ell)\right).$$

Hence

$$t = \frac{e^{-\mathrm{i}k\ell}}{\cosh(q\ell) + \mathrm{i}(q^2-k^2)\sinh(q\ell)/2kq}.$$

At $\ell = 0$ this gives $t = 1$, as it must without a barrier. For the values of Figure 11.2 it gives $t \approx -0.133 - 0.290\mathrm{i}$. The wave numbers on the two sides are equal, so the transmitted fraction of the current is $|t|^2$. Since $\cosh^2 = 1 + \sinh^2$, $(q^2-k^2)^2 + 4k^2q^2 = (q^2+k^2)^2$, $q^2 + k^2 = 2mV_0/\hbar^2$ and $k^2q^2 = 4m^2E(V_0 - E)/\hbar^4$, the transmission probability is

$$|t|^2 = \left[1 + \frac{V_0^2 \sinh^2(q\ell)}{4E(V_0 - E)}\right]^{-1}.$$

For $E > V_0$, the same matching, with q replaced by $\mathrm{i}q'$ and $q' = \sqrt{2m(E-V_0)}/\hbar$, gives

$$|t|^2 = \left[1 + \frac{V_0^2 \sin^2(q'\ell)}{4E(E - V_0)}\right]^{-1}.$$

For $E < V_0$ and $q\ell \gg 1$, $\sinh(q\ell) \approx e^{q\ell}/2$, and

$$|t|^2 \approx \frac{16E(V_0 - E)}{V_0^2}\, e^{-2q\ell}.$$

For the values of Figure 11.2, $q\ell \approx 1.79$, $|t|^2 \approx 0.102$, and the approximation gives about 0.107. The particle passes with a probability that falls exponentially with the width, at the rate $2q$ set by the missing energy $V_0 - E$. This is *tunnelling*. A transmitted particle does not lose energy in the barrier. Its wave on the right is $te^{\mathrm{i}kx}$, with the incoming k. Only the probability is reduced. Gamow explained α decay in this way in 1928. The α particle tunnels out through the Coulomb barrier of the nucleus. For a barrier of varying height, a semiclassical estimate, not derived here, gives a factor like $e^{-2q\ell}$. This factor spreads the lifetimes over more than twenty orders of magnitude. For example, the measured half-lives of uranium-238 and polonium-212 are about 4.5×10^9 years and 0.3 μs (Kondev et al., NUBASE2020).

Self-check 11.3

1. For $V_0 - E = 1$ eV and an electron, by what factor does $|t|^2$ fall when the barrier widens by 0.1 nm, for $q\ell \gg 1$?
2. Take $\hbar = m = V_0 = 1$, $E = 0.6$ and $\ell = 1$. Compute $|t|^2$ from the exact formula, and compare it with the approximation for $q\ell \gg 1$.

The physics

In 1982 Binnig, Rohrer, Gerber and Weibel reported measurements with a tungsten tip held a fraction of a nanometre from a platinum plate, with a gap they could set: the tunnelling current through the vacuum falls exponentially with the width

of the gap. The same year they scanned such a tip over a surface at a fixed current, tracing its height: the scanning tunnelling microscope. In the model of this section, an electron that lacks $V_0 - E = 4\,\mathrm{eV}$, of the order of the energy needed to free an electron from a metal, has $q \approx 10\,\mathrm{nm}^{-1}$, and a gap wider by 0.1 nm cuts the transmission by $e^{-2q\times 0.1\,\mathrm{nm}} \approx 1/8$.

11.4 The δ potential

The step and the barrier are bounded, so their Hamiltonians have the domain $D(H_0)$ and their solutions have continuous slopes. A narrow and deep well leads to a different model, in which the slope jumps.

Let $\alpha > 0$, and make the square well of § 10.4 narrow and deep with $\int V = -2wV_0 = -\alpha$ fixed. Then $\gamma^2 = m\alpha w/\hbar^2 \to 0$ as $w \to 0$, so for small w there is one level, on the first branch of Theorem 10.8. There $\eta w = kw \tan kw$. Since $\tan u = u + O(u^3)$ and $(kw)^2 = \gamma^2 - (\eta w)^2$, we get $\eta w = \gamma^2 + O(\gamma^4)$. So $\eta \to m\alpha/\hbar^2$, and the level tends to $-m\alpha^2/2\hbar^2$. Figure 11.3 draws the normalised ground states of three such wells. As w shrinks, they approach a function whose slope jumps at 0.

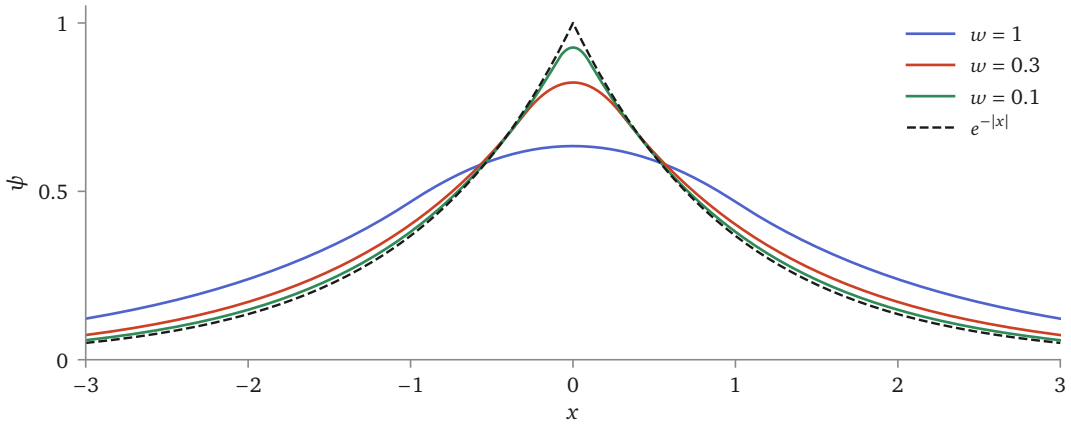


Figure 11.3: The normalised ground states of square wells with $\int V = -\alpha$ and the half-widths $w = 1$, 0.3 and 0.1, in units with $\hbar = m = \alpha = 1$, and dashed, the eigenvector $e^{-|x|}$ of the δ well (Theorem 11.5). The levels of the wells are about −0.23, −0.36 and −0.44, and that of the δ well is $-1/2$. Each well state has a continuous slope. The dashed limit has a corner at 0, where its slope jumps by $-2\psi(0)$, as in (11.2). The figure illustrates the convergence of the bound states. It follows from the norm-resolvent convergence cited after Theorem 11.5, and the figure does not prove it.

The limit is the *δ well* $-\alpha\delta$, an attraction at the single point 0. Formally, integrating $-(\hbar^2/2m)\psi'' - \alpha\delta\psi = E\psi$ over a small interval around 0 gives a jump of the slope at 0,

$$\psi'(0+) - \psi'(0-) = -\frac{2m\alpha}{\hbar^2}\,\psi(0). \tag{11.2}$$

This calculation is formal, since δ is not a function. The δ well is not a potential in the sense of Definition 10.1, and its Hamiltonian needs its own domain, of functions whose slope

jumps in this way. We build it from the free kernel. Fix $\eta > 0$, $z = -\hbar^2\eta^2/2m$ and the kernel G of Proposition 6.11. This kernel is continuous, its slope jumps at 0, and Lemma 11.4 below shows that $G \notin D(H_0)$. By Lemma 10.4, every $u \in D(H_0)$ is a continuous function, so $u(0)$ makes sense.

Definition 11.3 (The δ Hamiltonian)**.** Let α be real. The Hamiltonian H_α has the domain

$$D(H_\alpha) = \{\psi = u + cG : \ u \in D(H_0),\ c \in \mathbb{C},\ c = \alpha\psi(0)\},$$

and acts by $H_\alpha\psi = H_0 u + zcG$.

Lemma 11.4 (The domain)**.** *The decomposition $\psi = u + cG$ is unique, and $D(H_\alpha)$ and H_α do not depend on η.*

Proof. By Lemma 10.4, $\langle G, (H_0 - z)v\rangle = v(0)$ for every $v \in D(H_0)$, since G is real and even. If G lay in $D(H_0)$, then $v(0) = \langle (H_0 - z)G, v\rangle$, and $v(x) = f(nx)$, with f smooth, vanishing outside a bounded interval and $f(0) = 1$, would give $1 \le \|(H_0 - z)G\|\,\|f\|\,n^{-1/2}$ for every n. So $G \notin D(H_0)$, and the decomposition is unique. Let $\eta' > 0$, $z' = -\hbar^2\eta'^2/2m$, and write G_η, $G_{\eta'}$ for the two kernels. The two half-lines give the transform of G_η,

$$\tilde G_\eta(p) = \frac{1}{\sqrt{2\pi\hbar}}\,\frac{m}{\hbar^2\eta}\Big(\frac{1}{\eta + \mathrm{i}p/\hbar} + \frac{1}{\eta - \mathrm{i}p/\hbar}\Big) = \frac{1}{\sqrt{2\pi\hbar}}\,\frac{1}{p^2/2m - z}.$$

Since $H_0 = F^{-1}M_{p^2/2m}F$ (proof of Theorem 4.6), the operator $(H_0 - z')^{-1}$ multiplies transforms by $(p^2/2m - z')^{-1}$, as in the proof of Proposition 6.8. Since

$$\tilde G_{\eta'}(p) - \tilde G_\eta(p) = \frac{1}{\sqrt{2\pi\hbar}}\,\frac{z' - z}{(p^2/2m - z')(p^2/2m - z)},$$

we get $(z' - z)(H_0 - z')^{-1}G_\eta = G_{\eta'} - G_\eta$, so $G_\eta - G_{\eta'} \in D(H_0)$. With $u' = u + c(G_\eta - G_{\eta'})$, $\psi = u' + cG_{\eta'}$, and since $(H_0 - z')(G_{\eta'} - G_\eta) = (z' - z)G_\eta$, we get $H_0u' + z'cG_{\eta'} = H_0u + zcG_\eta$. ∎

Since $G'(0+) - G'(0-) = -2m/\hbar^2$, a function $\psi = u + cG$ of the domain with a continuously differentiable u is continuous, and its slope jumps at 0 by $-2mc/\hbar^2$. Since $c = \alpha\psi(0)$, this is the jump (11.2). For a smooth u, $H_\alpha\psi = -(\hbar^2/2m)\psi''$ away from 0, since $G'' = \eta^2 G$ there.

Theorem 11.5 (The δ potential)**.** *Let α be real. Then H_α is self-adjoint, and $\sigma_{\mathrm{ess}}(H_\alpha) = [0,\infty)$. If $\alpha \le 0$, then $\sigma(H_\alpha) = [0,\infty)$. If $\alpha > 0$, then H_α has exactly one eigenvalue,*

$$E = -\frac{m\alpha^2}{2\hbar^2},$$

which is simple and isolated, with the eigenvector $e^{-\mu|x|}$, $\mu = m\alpha/\hbar^2$, and $\sigma(H_\alpha) = \{E\} \cup [0,\infty)$.

Proof. First, H_α is symmetric. For $\psi_j = u_j + c_jG$ in $D(H_\alpha)$, $(H_\alpha - z)\psi_j = (H_0 - z)u_j$. With $\langle G, (H_0 - z)u_2\rangle = u_2(0)$, $u_2(0) = \psi_2(0) - c_2G(0)$ and $\overline{c_1} = \alpha\overline{\psi_1(0)}$,

$$\langle \psi_1, (H_\alpha - z)\psi_2\rangle = \langle u_1, (H_0 - z)u_2\rangle + \alpha\overline{\psi_1(0)}\psi_2(0) - G(0)\overline{c_1}c_2.$$

Since H_0 is symmetric and z, α and $G(0)$ are real, exchanging ψ_1 and ψ_2 gives the complex conjugate. So $\langle \psi_1, H_\alpha\psi_2\rangle = \langle H_\alpha\psi_1, \psi_2\rangle$. The domain is dense. It contains the smooth functions that vanish outside a bounded interval and near 0 (with $c = 0$). A smooth

φ that vanishes outside a bounded interval differs from such a function, $\varphi(1-\theta(x/\epsilon))$ with θ smooth, $0 \le \theta \le 1$, equal to 1 near 0 and vanishing outside $[-2,2]$, by at most $\sup|\varphi|\ (4\epsilon)^{1/2}$ in norm, and by S4 these φ are dense.

Next, let $\lambda = -\hbar^2\mu^2/2m < 0$, with $\mu > 0$, and write the domain with G_μ (Lemma 11.4). Then $(H_\alpha - \lambda)\psi = (H_0 - \lambda)u$, so $\psi = u + cG_\mu$ solves $(H_\alpha - \lambda)\psi = \varphi$ exactly when $u = (H_0-\lambda)^{-1}\varphi$ and $c = \alpha(u(0) + cG_\mu(0))$, that is,

$$\big(1 - \alpha G_\mu(0)\big)c = \alpha u(0). \tag{11.3}$$

If $\alpha G_\mu(0) \neq 1$, this has exactly one solution, and $|u(0)| \le \|G_\mu\|\,\|\varphi\|$ (Lemma 10.4), so ψ is a bounded function of φ and λ lies in the resolvent set. Since $\alpha G_\mu(0) = m\alpha/\hbar^2\mu$, this holds for every $\lambda < 0$ if $\alpha \le 0$, and for every $\lambda < 0$ but one if $\alpha > 0$. The proof of Proposition 6.7, with one such real λ in place of z and $\bar z$, shows that H_α is self-adjoint.

If $\alpha > 0$ and $\mu = m\alpha/\hbar^2$, then (11.3) holds with $u = 0$ and every c. So $G_\mu \in D(H_\alpha)$ and $H_\alpha G_\mu = \lambda G_\mu$. So the number $\lambda = -m\alpha^2/2\hbar^2$ is an eigenvalue, with an eigenvector G_μ, a multiple of $e^{-\mu|x|}$. Every eigenvector $u + cG_\mu$ has $(H_0 - \lambda)u = 0$, so $u = 0$, and the level is simple. The other negative numbers lie in the resolvent set, so it is isolated.

No eigenvalue is ≥ 0. Let $H_\alpha\psi = \lambda\psi$ with $\lambda \ge 0$ and $\psi = u + cG$. Then $(H_0 - z)u = (H_\alpha - z)\psi = (\lambda - z)\psi$, a continuous function, so by the second claim of Lemma 10.4 u is twice differentiable, with $-(\hbar^2/2m)u'' - zu = (\lambda - z)\psi$. Since $-(\hbar^2/2m)G'' = zG$ off 0, $-(\hbar^2/2m)\psi'' = \lambda\psi$ on each half-line, and as in the first paragraph of the proof of Theorem 10.7, a square-integrable solution vanishes there. So $\psi = 0$.

Finally, the packets ψ_n of the proof of Theorem 10.10, with $R = 1$, vanish near 0, so $\psi_n \in D(H_\alpha)$ with $c = 0$ and $H_\alpha\psi_n = H_0\psi_n$. As there, $[0,\infty) \subseteq \sigma_{\mathrm{ess}}(H_\alpha)$, and $\sigma(H_\alpha) \subseteq \mathbb{R}$ by Theorem 6.3. ∎

The bound state is the free kernel G at its own level, and the level condition reads $\alpha G(0) = 1$. The square wells converge to H_α in the norm-resolvent sense (Albeverio et al., Thm I.3.2.3): for each nonreal spectral parameter, the resolvents converge in operator norm.

With the other sign, the *δ barrier* $\alpha\delta$, with $\alpha > 0$, is $H_{-\alpha}$, with the jump $+(2m\alpha/\hbar^2)\psi(0)$. A wave $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ for $x < 0$ and $te^{\mathrm{i}kx}$ for $x > 0$ is continuous when $1 + r = t$, and has this jump when $\mathrm{i}kt - \mathrm{i}k(1-r) = (2m\alpha/\hbar^2)t$. So

$$t = \frac{1}{1 + \mathrm{i}m\alpha/\hbar^2 k}, \qquad r = t - 1, \qquad |t|^2 = \frac{1}{1 + m\alpha^2/2\hbar^2 E},$$

and $|r|^2 + |t|^2 = 1$. Slow particles are reflected and fast ones pass. The δ well has the same $|t|^2$.

Self-check 11.4

1. An electron sits in a δ well with $\alpha = 1\,\mathrm{eV\,nm}$. Find its level and the decay length $1/\eta$. Use $\hbar c = 197.3\,\mathrm{eV\,nm}$ and $mc^2 = 511\,\mathrm{keV}$.

So, at each energy above 0, a stationary solution gives the reflected and transmitted fractions of an incoming current: $|r|^2$ and $|t|^2$ for a potential that vanishes outside a bounded interval, and R and T for the step, whose current carries the velocity on each side. The δ potential needs its own domain, on which the slope jumps by (11.2).

Lecture 11, second part

The transmission of a packet

The first part read a stationary solution ψ_k as a stream of particles. A stream is not a state, since ψ_k is not square integrable. In this part we build the evolution of packets by a potential, compare it with the free evolution by wave operators, and obtain from the amplitudes r and t the probability that a packet is transmitted.

11.5 Packets and wave operators

A state is a packet, and it moves by the evolution of $H = H_0 + V$. In this section V is a potential (Definition 10.1): it is bounded, and it vanishes outside a bounded interval. The step does not vanish outside a bounded interval, and the δ Hamiltonian is not of the form $H_0 + V$ with a bounded V, so the results below do not cover them.

Theorem 7.5 builds that evolution with the spectral theorem. For a potential it can be built from the free evolution U_t^0 of (7.3) alone. We build it for two uses. The proof of Theorem 11.7 then needs no spectral theorem, and the construction also gives the evolution of the Friedrichs model in Lecture 12. If U_t were known, $W_t = U_{-t}^0 U_t$ would satisfy $\mathrm{i}\hbar\, \mathrm{d}W_t/\mathrm{d}t = B_t W_t$, with B_t below. We solve this equation instead.

Proposition 11.6 (The evolution with a potential). *Let V be a potential and $H = H_0 + V$, and put $B_s = U_{-s}^0 V U_s^0$. For every ψ there is exactly one continuous solution $t \mapsto W_t\psi$ of*

$$W_t\psi = \psi - \frac{\mathrm{i}}{\hbar}\int_0^t B_s W_s \psi \,\mathrm{d}s, \tag{11.4}$$

and $U_t = U_t^0 W_t$ is a strongly continuous unitary group with the generator $H/\hbar$. So for these H the group of Theorem 7.5 exists without the spectral theorem: it is $U_t = e^{-\mathrm{i}tH/\hbar}$.

Proof. Let $v = \sup|V|$. Then $\|B_s\| \le v$, and $B_s\chi$ is continuous in s for every χ, since U_s^0 is strongly continuous and keeps norms. So $B_s g(s)$ is continuous for every continuous g, since $\|B_{s'}g(s') - B_s g(s)\| \le v\,\|g(s') - g(s)\| + \|(B_{s'} - B_s)g(s)\|$.

Existence. Put $W_t^0\psi = \psi$, and let $W_t^{n+1}\psi$ be the right side of (11.4) with $W_s^n\psi$ in the integral. Each is continuous in t (Lemma 7.7), and by the norm bound of Lemma 7.7 and induction, $\left\|W_t^{n+1}\psi - W_t^n\psi\right\| \le \|\psi\|\,(v\,|t|\,/\hbar)^{n+1}/(n+1)!$. So the $W_t^n\psi$ converge uniformly for $|t| \le \tau$, for every τ, and the limit $W_t\psi$ is continuous, linear in ψ and bounded by $e^{v|t|/\hbar}\,\|\psi\|$. By the norm bound again, the integrals converge too, so $W_t\psi$ solves (11.4).

Uniqueness. Let a be real, and let χ be continuous with $\chi(t) = -(\mathrm{i}/\hbar)\int_0^t B_{s+a}\chi(s)\,\mathrm{d}s$. If M is the largest value of $\|\chi\|$ on $[-\tau,\tau]$, induction gives $\|\chi(t)\| \le M(v\,|t|\,/\hbar)^n/n!$ there, so $\chi = 0$. With $a = 0$, this is the uniqueness claimed.

Norms. Let $f(t) = W_t\psi$. By the last claim of Lemma 7.7, f is differentiable with $f'(t) = -(\mathrm{i}/\hbar)B_t f(t)$. So, by the product rule for $\langle f(t), f(t)\rangle$, the derivative of $\|f(t)\|^2$ is

$$2\,\mathrm{Re}\,\langle f(t), f'(t)\rangle = (2/\hbar)\,\mathrm{Im}\,\langle f(t), B_t f(t)\rangle = 0,$$

since B_t is symmetric. So $\|f(t)\|^2$ is constant, and $\|W_t\psi\| = \|W_0\psi\| = \|\psi\|$.

The group. Fix s, and put $\varphi = U_s^0 W_s\psi$. Since $U_{-s}^0 B_r = B_{r+s}U_{-s}^0$, splitting the integral for $W_{t+s}\psi$ at s shows that $\chi(t) = W_{t+s}\psi - U_{-s}^0 W_t\varphi$ satisfies the equation of the uniqueness step with $a = s$. So $\chi = 0$, that is, $U_{t+s}\psi = U_t U_s\psi$. Each U_t keeps norms and $U_t U_{-t} = U_0 = \mathbb{1}$, so U_t is unitary, and $U_t\psi$ is continuous in t, by the bound used for $B_s g(s)$.

The generator. For $\psi \in D(H_0)$,

$$\frac{U_h\psi - \psi}{h} = U_h^0 \frac{W_h\psi - \psi}{h} + \frac{U_h^0\psi - \psi}{h} \longrightarrow -\frac{\mathrm{i}}{\hbar}(V + H_0)\psi,$$

since $(W_h\psi - \psi)/h \to -(\mathrm{i}/\hbar)B_0\psi = -(\mathrm{i}/\hbar)V\psi$ by (11.4), and the U_h^0 keep norms and tend to $\mathbb{1}$ on every vector. So the generator extends $H/\hbar$, which is self-adjoint (Proposition 10.2), and as in the proof of Theorem 7.4 it equals $H/\hbar$. By Proposition 7.9, every group with the generator $H/\hbar$ is (U_t). ∎

The iterates W_t^n are the partial sums of the *Dyson series*. A packet that comes in from far away should move, at early times, like a free packet $U_t^0\psi$. The wave operators compare the two evolutions.

Theorem 11.7 (Wave operators). *Let V be a potential. For every $\psi \in L^2(\mathbb{R})$ the limits*

$$\Omega_{\mathrm{in}}\psi = \lim_{t\to-\infty} U_t^\dagger U_t^0\psi, \qquad \Omega_{\mathrm{out}}\psi = \lim_{t\to+\infty} U_t^\dagger U_t^0\psi$$

exist in norm, and $\|\Omega_{\mathrm{in}}\psi\| = \|\Omega_{\mathrm{out}}\psi\| = \|\psi\|$. Every eigenvector of H is orthogonal to $\Omega_{\mathrm{in}}\psi$ and to $\Omega_{\mathrm{out}}\psi$.

Proof. We treat Ω_{out}; Ω_{in} is the same with $t \to -\infty$. Write $f_\psi(t) = U_t^\dagger U_t^0\psi = U_{-t}U_t^0\psi$. Its derivative has the norm $\left\|VU_t^0\psi\right\|/\hbar$, so $f_\psi(t)$ converges if this norm is integrable in t. We prove the integrability, by a bound C/t^2, for smooth packets whose momentum support is compact and avoids 0. Density then gives every ψ.

Each $U_{-t}U_t^0$ is unitary, so $\left\|f_\psi(t) - f_\chi(t)\right\| = \|\psi - \chi\|$, and it suffices to prove convergence for ψ in a dense set. Let $0 < a < b$ and let $\tilde\psi$ be smooth and vanish outside $[a, b]$, or outside $[-b, -a]$. By S1 and S4, and the cut near 0 of the proof of Theorem 11.5, the sums of such ψ are dense.

Decay. By (7.3) and the inversion formula,

$$(U_t^0\psi)(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{\mathrm{i}\Phi(p)/\hbar}\tilde\psi(p)\,\mathrm{d}p, \qquad \Phi(p) = px - \frac{p^2 t}{2m}.$$

Let V vanish outside $[-R, R]$. For $|x| \le R$, $a \le p \le b$ and $|t| \ge 2mR/a$, $|\Phi'(p)| = |x - pt/m| \ge a\,|t|\,/2m$. Write $e^{\mathrm{i}\Phi/\hbar} = (\hbar/\mathrm{i}\Phi')(e^{\mathrm{i}\Phi/\hbar})'$ and integrate by parts twice, with no boundary terms. The integral becomes $-\hbar^2$ times the integral of $e^{\mathrm{i}\Phi/\hbar}$ multiplied by

$$\left(\frac{(\tilde\psi/\Phi')'}{\Phi'}\right)' = \frac{(\tilde\psi)''}{\Phi'^2} + \frac{3t(\tilde\psi)'}{m\Phi'^3} + \frac{3t^2\tilde\psi}{m^2\Phi'^4}.$$

Here $\Phi'' = -t/m$. The derivatives of $\tilde\psi$ are bounded on their fixed support. Since $|\Phi'| \ge a\,|t|\,/2m$, each term is bounded by a constant times $1/t^2$. So $\left|(U_t^0\psi)(x)\right| \le C/t^2$ on $[-R, R]$ for large $|t|$, and $\left\|VU_t^0\psi\right\| \le \sup|V|\ (2R)^{1/2}C/t^2$.

Cook's argument. The vector ψ lies in $D(H_0) = D(H)$, and so does $U_t^0\psi$ (Proposition 7.3). Divide

$$f_\psi(t+h) - f_\psi(t) = U_{-t-h}\big(U_{t+h}^0\psi - U_t^0\psi\big) + \big(U_{-t-h} - U_{-t}\big)U_t^0\psi$$

by h. Since $\|U_{-t-h}\| = 1$ and U_{-t-h} tends to U_{-t} on every vector, the first term tends to $-(\mathrm{i}/\hbar)U_{-t}H_0U_t^0\psi$, and by Propositions 7.3 and 11.6 the second to $(\mathrm{i}/\hbar)U_{-t}HU_t^0\psi$. So

$f'_\psi(t) = (\mathrm{i}/\hbar)U_{-t}VU_t^0\psi$, a continuous function of t. For every φ, the derivative of $\langle\varphi, f_\psi(t)\rangle$ is $\langle\varphi, f'_\psi(t)\rangle$. Integrating it, with $\varphi = f_\psi(t') - f_\psi(t)$, and dividing by $\|\varphi\|$,

$$\|f_\psi(t') - f_\psi(t)\| \le \frac{1}{\hbar}\int_t^{t'} \|VU_s^0\psi\|\,\mathrm{d}s \le \frac{\sup|V|\ (2R)^{1/2}C}{\hbar}\left(\frac{1}{t} - \frac{1}{t'}\right)$$

for large $t < t'$. So $f_\psi(t)$ converges as $t \to \infty$. Its norm is $\|\psi\|$ for every t, and so is the norm of the limit.

Bound states. Let $H\varphi = E\varphi$. By Propositions 7.3 and 11.6, $e^{\mathrm{i}Et/\hbar}U_t\varphi$ has the derivative 0, so it is constant and $U_t\varphi = e^{-\mathrm{i}Et/\hbar}\varphi$. Hence $|\langle\varphi, f_\psi(t)\rangle| = |\langle\varphi, U_t^0\psi\rangle|$. For $L \ge R$ the decay step on $[-L, L]$ bounds this by $\|1_{|x|>L}\varphi\|\,\|\psi\| + (2L)^{1/2}\,\|\varphi\|\,C_L/t^2$, which is small for large L and then large t. So $\langle\varphi, \Omega_{\mathrm{out}}\psi\rangle = 0$ for these ψ, and by density for all. ∎

This is Cook's method (Teschl, Lemma 12.3).

Since U_t is unitary,

$$\|U_t\Omega_{\mathrm{out}}\psi - U_t^0\psi\| = \|\Omega_{\mathrm{out}}\psi - U_t^\dagger U_t^0\psi\| \longrightarrow 0.$$

So the state $\Omega_{\mathrm{out}}\psi$ moves at late times like the free packet $U_t^0\psi$, and $\Omega_{\mathrm{in}}\psi$ at early times. These are states that leave, and the theorem shows that they are orthogonal to the bound states, which stay. More is true, and cited. By Enss's theorem (Teschl, Thm 12.12), every state orthogonal to all bound states is some $\Omega_{\mathrm{in}}\psi$ and some $\Omega_{\mathrm{out}}\chi$: *asymptotic completeness*. So the *scattering operator* $S = \Omega_{\mathrm{out}}^\dagger\Omega_{\mathrm{in}}$ is unitary and maps the incoming free packet ψ to the outgoing one, $\psi_{\mathrm{out}} = S\psi$. It commutes with H_0 (Teschl, (12.10)), so the kinetic energy is kept. This is the packet form of what the stationary solution of § 11.3 shows, with the same k on both sides of the barrier.

An incoming free packet φ gives one actual state $\Omega_{\mathrm{in}}\varphi$. At early times it moves like $U_t^0\varphi$, and, as shown after the next theorem, at late times like $U_t^0 S\varphi$. Figure 11.4 shows this.

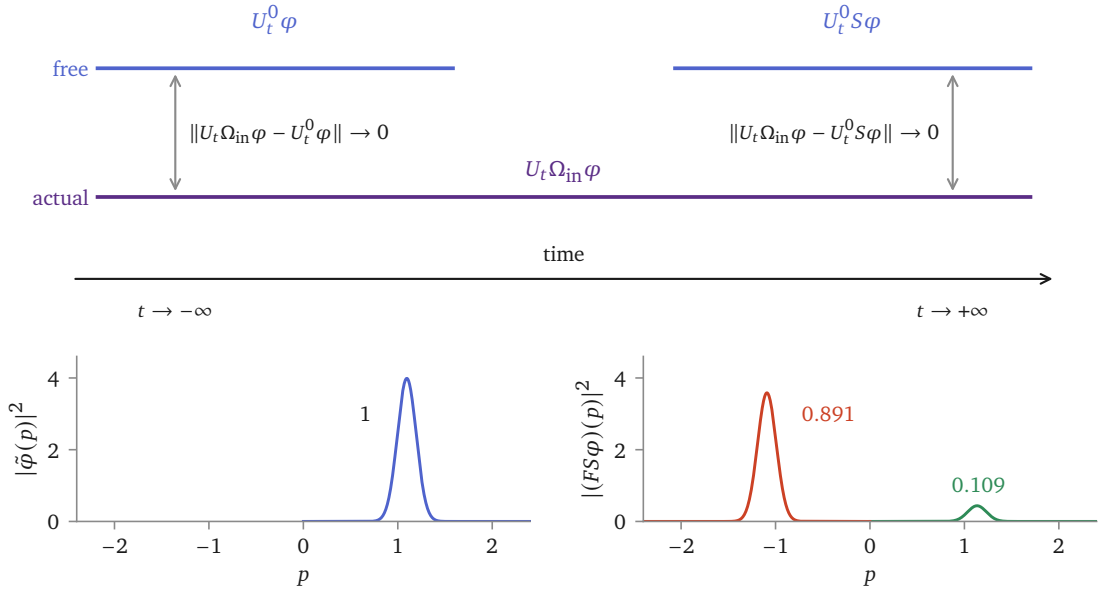


Figure 11.4: One actual state and its two free representatives. Upper part: the state $U_t\Omega_{\mathrm{in}}\varphi$ on one line, and on the other the free packets that it approaches in norm as $t \to -\infty$ and as $t \to +\infty$. No trajectory is computed: these two limits are what is drawn. Lower part: the momentum densities of φ and of $S\varphi$ by Theorem 11.8, for the barrier of § 11.3 ($\hbar = m = V_0 = 1$, $\ell = 2$) and the Gaussian of Figure 11.5 restricted to $p > 0$. The numbers are the masses under the curves, and those on the right sum to 1. The reflected peak lies at a smaller $|p|$ than the transmitted one, since $|t(p/\hbar)|^2$ grows with p there.

The stationary amplitudes enter through a superposition. The following argument is a heuristic, not a proof. The amplitudes carry their argument, $t(p/\hbar)$ and $r(p/\hbar)$, to keep them apart from the time t. Let V vanish outside $[-R, R]$, and let φ be a unit vector whose transform $\tilde{\varphi}$ is smooth and vanishes outside a closed bounded interval of $(0, \infty)$. We superpose the scattering solutions of Proposition 11.2, each with its own time factor:

$$\Psi_t(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_0^\infty \tilde{\varphi}(p)\, \psi_{p/\hbar}(x)\, e^{-\mathrm{i}p^2 t/2m\hbar}\, \mathrm{d}p.$$

Each $\psi_{p/\hbar}(x)\, e^{-\mathrm{i}p^2 t/2m\hbar}$ solves $\mathrm{i}\hbar\, \partial_t \Psi = -(\hbar^2/2m)\Psi'' + V\Psi$ where V is continuous, and so, formally, does Ψ_t. The incoming wave $e^{\mathrm{i}kx}$ of ψ_k has the amplitude 1. We define two packets χ_T and χ_R by their transforms. For $p > 0$ they are

$$\tilde{\chi}_T(p) = t(p/\hbar)\, \tilde{\varphi}(p) \quad \text{and} \quad \tilde{\chi}_R(-p) = r(p/\hbar)\, \tilde{\varphi}(p),$$

and $\tilde{\chi}_T = 0$ on $(-\infty, 0)$, $\tilde{\chi}_R = 0$ on $(0, \infty)$. By (11.1), (7.3) and the inversion formula, with p replaced by $-p$ in the reflected term,

$$\Psi_t = U_t^0 \chi_T \ \text{ on } [R, \infty), \qquad \Psi_t = U_t^0 \varphi + U_t^0 \chi_R \ \text{ on } (-\infty, -R].$$

The packets φ and χ_T have positive momenta and move to the right. The packet χ_R has negative momenta and moves to the left. At early times, $U_t^0\varphi$ and $U_t^0\chi_T$ lie far to the left, and $U_t^0\chi_R$ lies far to the right. So only $U_t^0\varphi$ lies where its formula holds, and Ψ_t is nearly the

incoming free packet. At late times, $U_t^0\varphi$ and $U_t^0\chi_T$ lie far to the right, and $U_t^0\chi_R$ lies far to the left. Then $U_t^0\varphi$, which enters only the formula on the left, is negligible there, and Ψ_t is nearly $U_t^0\chi_T$ on the right and $U_t^0\chi_R$ on the left. Here a packet lies far to one side when nearly all its probability lies there. It need not vanish elsewhere. Suppose that Ψ_t is the actual state $U_t\Omega_{\mathrm{in}}\varphi$, which this argument does not show. Then it moves at late times like $U_t^0(\chi_T+\chi_R)$, so $S\varphi=\chi_T+\chi_R$, and its probability at $x>0$ tends to $\|\chi_T\|^2=\int_0^\infty |t(p/\hbar)|^2\,|\tilde\varphi(p)|^2\,\mathrm{d}p$. The next theorem states this for every unit packet incoming from the left.

Theorem 11.8 (Transmission of a packet · *sketched*)**.** *Let V be a potential, and let φ be a unit vector with $\tilde\varphi=0$ on $(-\infty,0)$, a packet incoming from the left. For $p>0$,*

$$(FS\varphi)(p)=t(p/\hbar)\,\tilde\varphi(p)\quad\text{and}\quad(FS\varphi)(-p)=r(p/\hbar)\,\tilde\varphi(p).$$

The probability that $U_t\Omega_{\mathrm{in}}\varphi$ is read at $x>0$ tends, as $t\to\infty$, to

$$\int_0^\infty |t(p/\hbar)|^2\,|\tilde\varphi(p)|^2\,\mathrm{d}p.$$

The action of S is Takhtajan, Thm 2.5. The limit follows from it. By asymptotic completeness, $\Omega_{\mathrm{in}}\varphi=\Omega_{\mathrm{out}}S\varphi$, so $U_t\Omega_{\mathrm{in}}\varphi-U_t^0S\varphi\to0$ in norm, and the two states have the same limiting probabilities. We use the late-time law of free packets, a cited theorem that Lecture 12 states in full (Theorem 12.5). Applied to the interval $(0,\infty)$, it says that for every unit vector χ the probability that $U_t^0\chi$ is read at $x>0$ tends, as $t\to\infty$, to the probability of positive momentum in χ, which is $\int_0^\infty|\tilde\chi(p)|^2\,\mathrm{d}p$. With $\chi=S\varphi$, this is the integral above.

Teta, (8.79)–(8.81), finds the same integral for a Gaussian packet far to the left of a δ barrier, up to an exponentially small error. So for an incoming packet whose momenta lie so near $\hbar k$ that $|t(p/\hbar)|^2$ hardly varies on them, $|t(k)|^2$ is approximately its transmission probability.

Figure 11.5 computes one case, for the barrier of § 11.3 in units with $\hbar=m=V_0=1$ and $\ell=2$. The packet starts as

$$\psi_0(x)=(2\pi s^2)^{-1/4}e^{-(x-x_0)^2/4s^2+\mathrm{i}p_0x},\qquad s=5,\quad x_0=-40,\quad p_0=\sqrt{1.2},$$

and evolves until $t=70$. It is a packet of mean energy about $0.6V_0$: precisely, $p_0^2/2+1/8s^2=0.605$. At the central energy $p_0^2/2=0.6$, $|t(k)|^2\approx0.102$. About 11% of the packet passes. This is the mean of $|t(k)|^2$ under its distribution of energies $E=\hbar^2k^2/2m$, which is about 0.1095. The momentum density of this Gaussian has a tail on $p<0$, of mass about 3×10^{-28}, so it satisfies the hypothesis of Theorem 11.8 only approximately. The tail is negligible, not zero. We also use ψ_0 in place of $\Omega_{\mathrm{in}}\psi_0$ and $t=70$ in place of the limit. The reason for the first is that the centre of the free packet starts far to the left and, going back in time, moves farther left, where $V=0$. We do not estimate these two differences.

Figure 11.6 shows the same computation at every time between the start and $t=70$, which Figure 11.5 shows only at its two ends. The packet splits at the barrier into a reflected part and a transmitted part. While the incoming and the reflected parts overlap, they interfere, and the density has fringes.

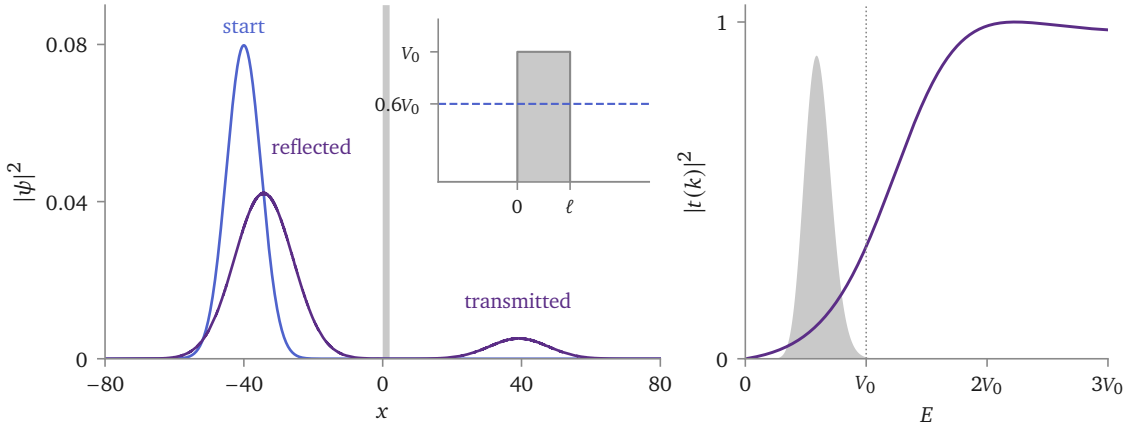


Figure 11.5: A packet at the barrier of § 11.3, of height V_0 and width ℓ, with $mV_0\ell^2/\hbar^2 = 4$. Left, in units with $\hbar = m = V_0 = 1$: the position density at the start and at the later time $t = 70$, computed numerically by the split-step Fourier method; the barrier is the thin grey strip at 0. The inset shows the barrier at a larger scale, with the central energy dashed. Right: $|t(k)|^2$ against the energy $E = \hbar^2k^2/2m$, and the packet's distribution of energies (grey, not to scale). The transmitted probability, 0.11, is the mean of $|t(k)|^2$ under it.

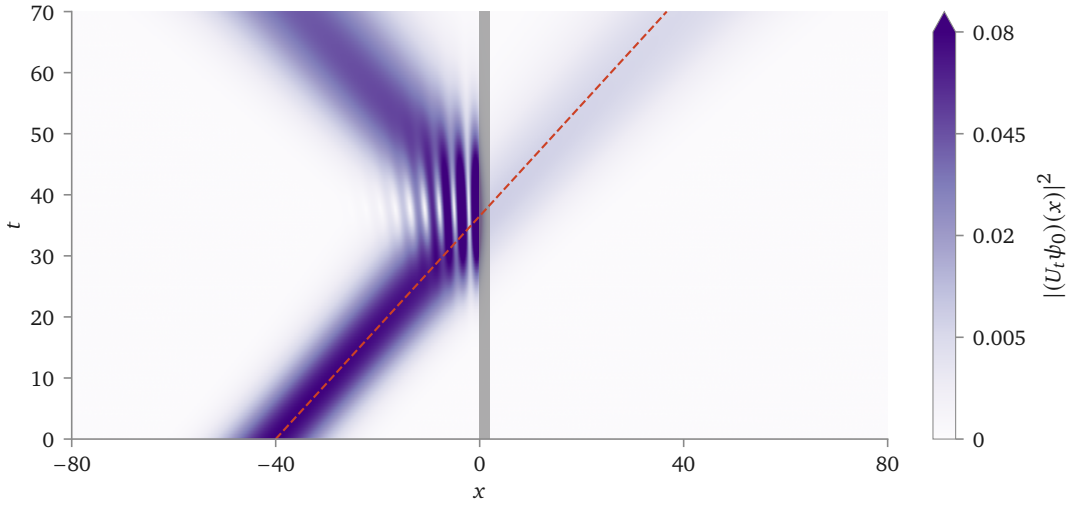


Figure 11.6: The position density $|(U_t\psi_0)(x)|^2$ of Figure 11.5 against x (horizontal) and t (vertical), from the same split-step computation, with the barrier grey. The colour follows the square root of the density, so that the transmitted part stays visible, and the key stops at 0.08, which the fringes exceed. The dashed line is $x_0 + p_0t/m$, the path of the centre of the free packet $U_t^0\psi_0$, without the barrier. It is not a classical path: a classical particle with the central energy $0.6V_0$ is reflected.

Self-check 11.5

1. Show that $\Omega_{\text{out}} U_s^0 = U_s \Omega_{\text{out}}$ for every real s.
2. Which of the step of § 11.2, the barrier of § 11.3 and the δ barrier of § 11.4 does Theorem 11.8 cover? Say why it does not cover the others.
3. Take the barrier in units with $\hbar = m = V_0 = 1$ and $\ell = 2$. A unit incoming packet φ, with $\tilde{\varphi} = 0$ on $(-\infty, 0)$, has half of its momentum probability in a narrow band near $p = 1$ and half in a narrow band near $p = 2$. Estimate the probability that it is read at $x > 0$ at late times, and compare it with $|t(k)|^2$ at the mean momentum $\hbar k = 3/2$.

Scattering

States: the scattering solutions ψ_k (Proposition 11.2) are not square integrable, so they are not states; packets are. For a potential V (Definition 10.1), the state $\Omega_{\text{out}}\psi$ moves at late times like the free packet $U_t^0\psi$, and it is orthogonal to the bound states (Theorem 11.7). *Observables:* the δ Hamiltonian H_α is self-adjoint on $D(H_\alpha)$, with $\sigma_{\text{ess}}(H_\alpha) = [0, \infty)$ and one level $-m\alpha^2/2\hbar^2$ for $\alpha > 0$ (Theorem 11.5). *Measurement:* by Theorem 11.8, for a potential and a unit incoming free packet φ with $\tilde{\varphi} = 0$ on $(-\infty, 0)$, the state $U_t \Omega_{\text{in}} \varphi$ is read at $x > 0$, as $t \to \infty$, with probability tending to $\int_0^\infty |t(p/\hbar)|^2 \, |\tilde{\varphi}(p)|^2 \, \mathrm{d}p$.

Summary.

- *Scattering solutions*: for a potential that vanishes outside $[-R, R]$ and $k > 0$, the equation has exactly one solution $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ on the left and $te^{\mathrm{i}kx}$ on the right, and $|r|^2 + |t|^2 = 1$ (Lemma 11.1 and Proposition 11.2).
- *The step*: $R = ((k - k')/(k + k'))^2$ and $T = 4kk'/(k + k')^2$ above it, $R = 1$ below it.
- *Tunnelling*: a barrier transmits about $16E(V_0 - E)V_0^{-2}e^{-2q\ell}$ for $q\ell \gg 1$, with no loss of energy.
- *The δ potential*: the Hamiltonian H_α of $-\alpha\delta$ is self-adjoint on $D(H_\alpha)$, with one level $-m\alpha^2/2\hbar^2$ if $\alpha > 0$ (Theorem 11.5); the barrier $\alpha\delta$ transmits $|t|^2 = 1/(1 + m\alpha^2/2\hbar^2 E)$.
- *Evolution and wave operators*: for a potential, the Dyson series builds $e^{-\mathrm{i}tH/\hbar}$ (Proposition 11.6), and Ω_{in} and Ω_{out} exist, keep norms and miss the bound states (Theorem 11.7); asymptotic completeness and S are cited, and they give the packet's transmission limit (Theorem 11.8).

Lecture 12 asks what the states that stay and the states that leave do in time.

Exercises. Exercise Sheet 11.

Self-check answers.

11.1 1. Expanding $(e^{\mathrm{i}kx} + re^{-\mathrm{i}kx})(e^{-\mathrm{i}kx} + \bar{r}e^{\mathrm{i}kx})$ gives $1 + |r|^2 + re^{-2\mathrm{i}kx} + \overline{re^{-2\mathrm{i}kx}}$. As x varies, the last two terms take every value $2|r|\cos\theta$, so $|\psi_k|^2$ runs between $(1 - |r|)^2$ and $(1 + |r|)^2$.

11.2 1. With $s = k'/k = \sqrt{1 - V_0/E}$, $T = 4s/(1 + s)^2$. As E falls to V_0, $s \to 0$ and $T \to 0$; as $E \to \infty$, $s \to 1$ and $T \to 1$. 2. Here $k'/k = 1/2$. The equations $1 + r = t$ and $k(1 - r) = k't$ give $r = 1/3$ and $t = 4/3$. So $R = r^2 = 1/9$ and $T = t^2/2 = 8/9$, and $R + T = 1$. But $r^2 + t^2 = 17/9$, since the current weights $|t|^2$ by the velocity.

11.3 1. $q = \sqrt{2mc^2 \times 1\,\text{eV}}/\hbar c \approx 5.12\,\text{nm}^{-1}$, so the factor is $e^{-2q \times 0.1\,\text{nm}} \approx 0.36$, about $1/2.8$. 2. Here $q = \sqrt{0.8} \approx 0.894$, so the exact formula gives $|t|^2 = [1 + \sinh^2(0.894)/0.96]^{-1} \approx 0.481$, and the approximation gives $3.84\,e^{-1.789} \approx 0.642$. At $q\ell \approx 0.89$ the approximation is poor, since $q\ell$ is not large.

11.4 1. $E = -mc^2\alpha^2/2(\hbar c)^2 = -511\,000/(2\times 197.3^2)\,\mathrm{eV} \approx -6.56\,\mathrm{eV}$, and $1/\eta = (\hbar c)^2/(mc^2\alpha) = 197.3^2/511\,000\,\mathrm{nm} \approx 0.076\,\mathrm{nm}$.

11.5 1. Since $U_t^\dagger = U_{-t} = U_s U_{-t-s}$, $U_t^\dagger U_t^0 U_s^0 \psi = U_s U_{t+s}^\dagger U_{t+s}^0 \psi$. As $t \to \infty$, the left side tends to $\Omega_{\mathrm{out}} U_s^0 \psi$, and the right side to $U_s \Omega_{\mathrm{out}} \psi$, since U_s is bounded. 2. The theorem covers only the barrier. It is real and bounded, continuous except at its two jumps, and vanishes outside $[0, \ell]$, so it is a potential (Definition 10.1). The step does not vanish outside a bounded interval, and the δ Hamiltonian is not of the form $H_0 + V$ with a bounded V. 3. By Theorem 11.8 the probability tends to $\int_0^\infty |t(p/\hbar)|^2\, |\tilde{\varphi}(p)|^2 \, \mathrm{d}p$. On each narrow band $|t(p/\hbar)|^2$ hardly varies, so the integral is close to half the sum of its values at $p = 1$ and $p = 2$. At $p = 1$, $E = 1/2 < V_0$ and $q = 1$, so $|t|^2 = [1 + \sinh^2(2)]^{-1} \approx 0.071$. At $p = 2$, $E = 2 > V_0$ and $q' = \sqrt{2}$, so $|t|^2 = [1 + \sin^2(2\sqrt{2})/8]^{-1} \approx 0.988$. The probability is about 0.53. At $\hbar k = 3/2$, $E = 9/8$ and $q' = 1/2$, so $|t(k)|^2 = [1 + (16/9)\sin^2(1)]^{-1} \approx 0.443$. The transmission of a packet is the mean of $|t|^2$ under its distribution of momenta, not $|t|^2$ at its mean momentum.

References

Albeverio, Gesztesy, Høegh-Krohn and Holden, *Solvable Models in Quantum Mechanics*, Thm I.3.1.1 (p. 76), Thms I.3.1.3, I.3.1.4 (p. 78), Thm I.3.2.3 (p. 81). They describe the domain of Definition 11.3 (Thm I.3.1.3) and state Theorem 11.5 (Thms I.3.1.1 and I.3.1.4), for their $-\Delta_{\alpha',0}$ in units with $\hbar^2/2m = 1$ and $\alpha' = -2m\alpha/\hbar^2$. Teschl, *Mathematical Methods in Quantum Mechanics*, (12.10) (p. 248), Lemma 12.3 (p. 249), Thm 12.12 (p. 256). Takhtajan, *Quantum Mechanics for Mathematicians*, Thm 2.5 (p. 175). Teta, *A Mathematical Primer on Quantum Mechanics*, (8.79)–(8.81) (p. 190). Binnig, Rohrer, Gerber and Weibel, *Appl. Phys. Lett.* 40, 178 (1982), and *Phys. Rev. Lett.* 49, 57 (1982). Gamow, *Z. Phys.* 51, 204 (1928). F. G. Kondev, M. Wang, W. J. Huang, S. Naimi and G. Audi, The NUBASE2020 evaluation of nuclear physics properties, *Chin. Phys. C* 45, 030001 (2021).